



Janani Rangila

IT Undergraduate

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TECHNICAL SKILLS

- Programming:
Python, Dart (Flutter), Java, HTML/CSS, JavaScript, TypeScript
- Mobile App Development:
Flutter, Java, Firebase
- Web App Development:
React, TypeScript, MySQL
- UX/UI Design:
Figma, Inkscape

TECH STACK

- Visual Studio
- Android Studio Google
- Colab Firebase, SQLite
- AdobeXD,PS,Premire Pro
- Inkscape
- Figma

SOFT SKILLS

- Management Skills
- Problem-Solving
- Teamwork
Collaboration
- Critical Thinking
- Communication Skills
- Time Management

PROFILE

Highly motivated IT undergraduate with a strong interest in AI, blockchain, and IoT. Experienced in developing full-stack applications. Proficient in Python, FastAPI, Vue 3, and TypeScript. Skilled in system design, building scalable, real-world solutions.

PROJECTS

Food Delivery Mobile Application

- **Purpose:** To build a cross-platform food delivery app with user login, menu browsing, cart, and checkout features, and cloud database support for rural and small restaurants
- **Novelty:** Lightweight and cost-effective food delivery solution designed for rural and small-scale restaurants
- **My Contribution:** UI design, order management, database integration, frontend development, and feature implementation
- **Tech stack:** Flutter, Dart, and Firebase

Weather Web Application

- **Purpose:** Development of a responsive weather web application with real-time weather data retrieval, geolocation-based updates, and API-driven frontend integration
- **Novelty:** Real-time weather application with geolocation-based updates and dark/light mode support
- **My Contribution:** UI design, API integration, state management, and responsive frontend development
- **Tech stack:** React, Tailwind CSS, OpenWeatherMap API, and JavaScript

Student Course Registration System

- **Purpose:** Development of a secure web-based course registration system with login, role-based access (student and admin), course enrollment, and database management for academic records
- **Novelty:** Secure course registration system with separate student and admin management panels
- **My Contribution:** Frontend and backend development, database management, authentication, and validation feature implementation
- **Tech stack:** PHP, MySQL, Apache, HTML, CSS, and JavaScript (LAMP Stack)

RESEARCH PUBLICATIONS

- [1] **J. Rangila**, A. Siriweera, I. Paik, K. Naruse, I. Jayanada, & V. Devindi, “S5-SHB Agent: Society 5.0 enabled Multi-model Agentic Blockchain Framework for Smart Home,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.05027>
- **Purpose:** Development of a smart home system that uses multiple AI agents to coordinate devices, ensure secure IoT communication, maintain data integrity using blockchain, and enable real-time device interaction through MQTT and automation.
 - **Novelty:** Society 5.0 governance enabled an Agentic-blockchain framework for a secure and autonomous smart home IoT communication
 - **My Contribution:** System modeling and manuscript preparation
 - **Tech stack:** Python, Blockchain, MQTT, Agentic AI, and IoT simulation technologies

EDUCATION

G.C.E.Advance Level

- Ferguson High School Rathnapura 2018-2020

BSc(Hons) information technology

- Kotelawala Defence University (2023-Present)

[2] A. Siriweera, **J. Rangila**, K. Naruse, I. Paik, and I. Jayanada, “S5-HES Agent: Society 5.0-driven Agentic Framework to Democratize Smart Home Environment Simulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.01554> (*Under review, IEEE Open Journal of the Computer Society*)

- **Purpose:** Development of a smart home simulation framework requiring multi-agent AI coordination, IoT data simulation, RAG-based knowledge retrieval, and integration of multiple LLMs for automated and scalable testing.
- **Novelty:** Society 5.0-driven Agentic AI smart home simulation framework with RAG-based automation and multi-LLM integration for scalable IoT testing
- **My Contribution:** A member of the development team, System modeling, and manuscript preparation
- **Tech stack:** Python, MQTT, RAG, REST APIs, IoT simulation technologies, and Multi-LLM integration

[3] **J. Rangila**, A. Siriweera, K. Naruse, and I. Paik, “Case Study on Agentic AI-enabled MultiChain 2 Community-driven IoT Ecosystem,” 2026 International Conference on Electronics, Information, and Communication (ICEIC), Macau, China, 2026, pp. 1–5, doi: 10.1109/ICEIC69189.2026.11386375

- **Purpose:** Development of a healthcare IoT ecosystem requiring secure real-time data sharing, blockchain-based data integrity, MQTT-based IoT communication, and agentic AI for autonomous monitoring and management of medical data streams
- **Novelty:** Agentic AI-enabled MultiChain 2 healthcare IoT ecosystem enabling secure, real-time, and autonomous medical data management with intelligent AI agents
- **My Contribution:** A member of the development team, System modeling, and manuscript preparation
- **Tech stack:** Python, MultiChain 2, MQTT, Blockchain, Agentic AI, IoT technologies, and REST APIs

REFERENCES

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